

U-Net Segmentation of Canine Cutaneous Tumours

Weeks 3–4 Progress — Model Development (Phase 2 start)

Muhammad Tayyab Abbas · MSc AI & Data Science · University of Hull
Supervisor: Dr Claire Cashmore · 2nd fortnightly meeting

What was built

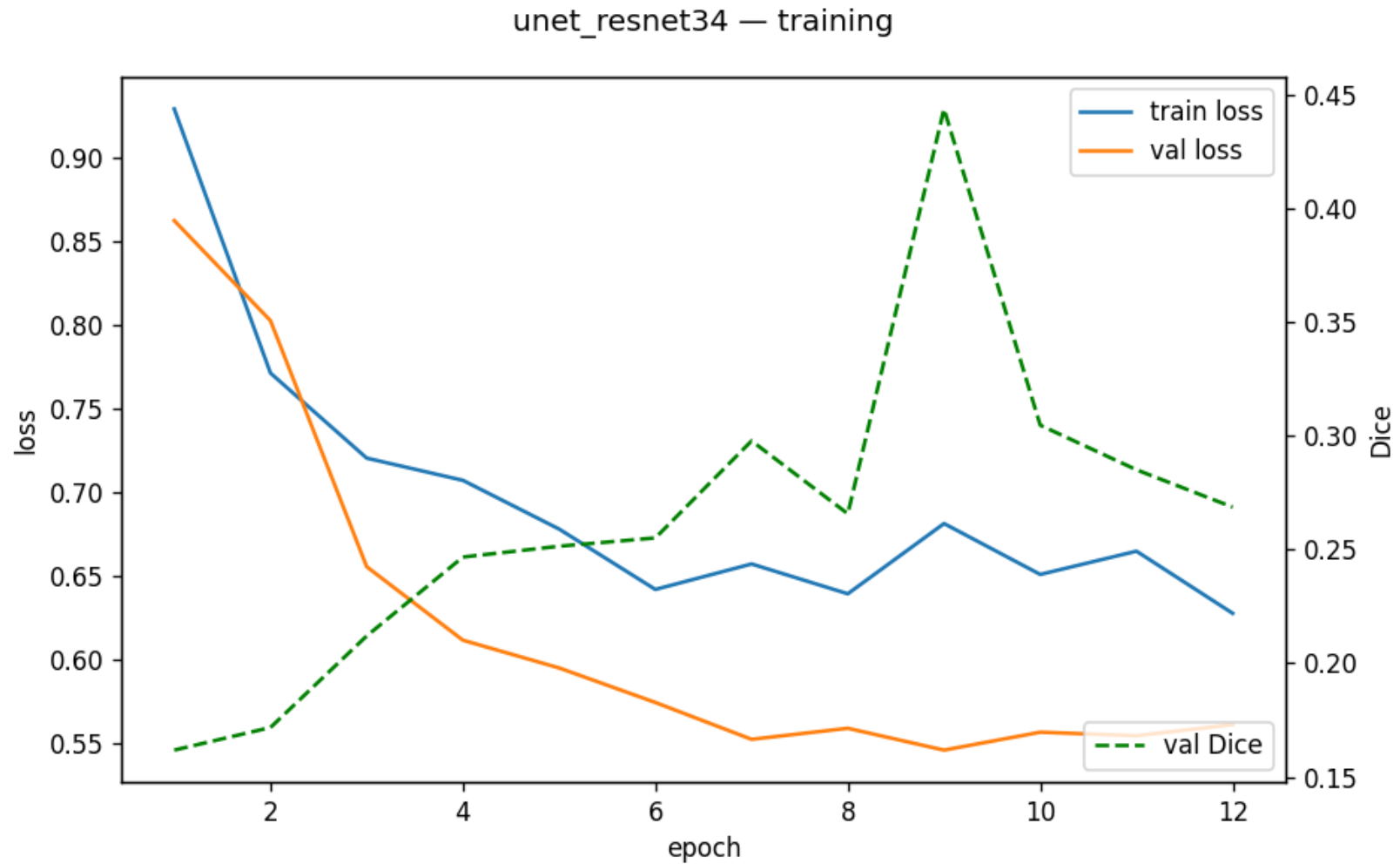
- Finished Phase-1 preprocessing: 5x/10x/20x multi-magnification patches + augmentation
- U-Net segmentation: from-scratch baseline → ResNet-34 encoder (transfer learning)
- BCE+Dice loss, Adam 1e-4, cosine annealing, early stopping, TensorBoard
- Slide-level split (no patch leakage); Dice/IoU/pixel-acc/Hausdorff evaluation

Results — baseline vs ResNet-34 (test)

Model	Dice	IoU	Pixel acc.	Hausdorff
Baseline U-Net	0.218	0.204	0.993	154.8
U-Net + ResNet-34	0.430	0.405	0.991	115.1

Synthetic-data validation of the full pipeline; proposal target is Dice > 0.85 on real CATCH slides.

Training curves (ResNet-34 U-Net)



Loss decreases and validation Dice rises — correct convergence.

Test predictions

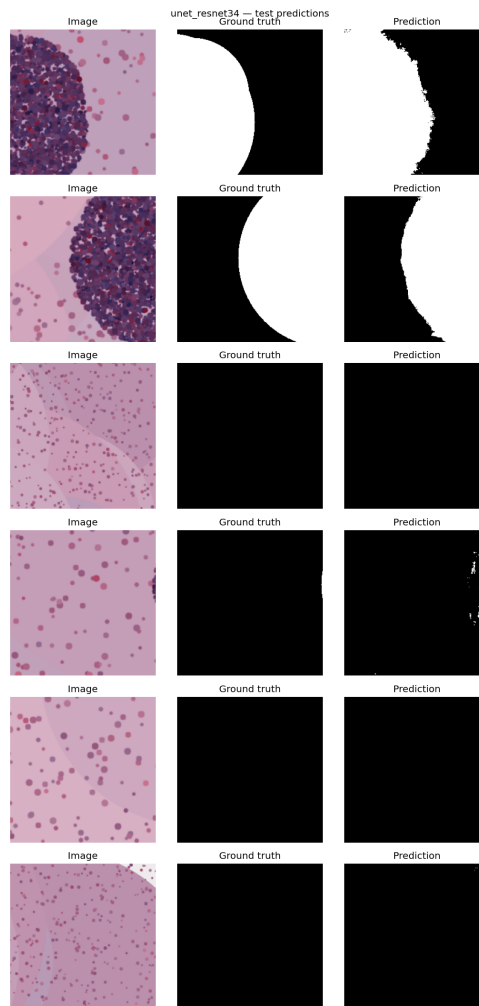


Image | ground-truth mask | predicted mask.

Next fortnight (Weeks 5–6)

- Attention U-Net variant + comparison
- Run on real CATCH annotated slides once downloaded
- Begin ResNet-50 classification

Thank you

Questions & feedback welcome